Choosing the Next Experiment in a Multi-Step Powder-to-Part Workflow, with Printable High-Strength Aluminum Alloys as the Test Case

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Research question. Developing a new alloy for laser powder bed fusion (LPBF) runs through a chain of steps, from a thermodynamic calculation to powder, printed tracks and cubes, microscopy, and tensile tests. A calculation takes minutes and a tensile test takes days. Campaigns often send every candidate through the whole chain, or screen many by simulation and test a few. We ask how to decide, as data arrive, which step of a multi-step manufacturing workflow to run next and on which candidate, while carrying uncertainty through the chain. Two features make this harder: several properties compete, and results from many experiments in progress return at different times. Our test case is high-strength aluminum, since alloys such as 7075 crack as they solidify during LPBF. Grain-refining additions such as zirconium can prevent the cracking (Martin et al., *Nature* 2017), but each change in composition means the laser settings must be tuned again.

Approach. We will give each step its own statistical model and link the models in order, so the outputs of one step feed the next. Recent work on such linked models picks the experiment with the largest expected gain per unit cost and lets a candidate stop at any step (Astudillo and Frazier, NeurIPS 2021; Buathong et al., ICML 2024). It has been shown for one objective and one experiment at a time, on test functions and simulators. We will extend it to weigh competing objectives (freedom from cracks and pores, strength, and ductility) with no weighting known in advance, and to choose new experiments while others are pending, since microscopy results arrive days after printing. We will demonstrate it on the five-step workflow, with real build-to-build scatter and with incomplete physics inputs from calculation of phase diagrams (CALPHAD), which ignores the fast cooling of the melt pool. We will test whether the method reaches a target set of trade-offs at lower total cost than standard multi-objective Bayesian optimization run on the full chain. We will also test whether its predicted chance of a dense, crack-free build matches observed rates on held-out builds.

Resources and preliminary results. Our laboratory atomizes aluminum alloy powder (AMAZEMET re-POWDER ultrasonic atomizer, induction melting), four compositions per week. We buy LPBF builds from a University of Utah user facility ($800 per build) and use the BYU Microscopy Center. Since the lab started in December 2025, we have run CALPHAD on BYU high-performance computing and completed a powder-to-part demonstration on a baseline aluminum alloy, with microscopy and tensile data. In a hand-built, noise-free simulation with one objective, where the CALPHAD crack screen missed zirconium's effect, screening and then tuning stalled at 63% of the optimum after $100,000. The cost-aware search reached that level after about $26,500 (3.8 times less spend; 90% confidence interval 2.0 to 5.7). Screening kept pace when its physics was right.

Intellectual merit. To our knowledge, neither a multi-objective nor an asynchronous version of cost-aware optimization over linked steps has been developed, and none has been tested on a physical manufacturing workflow. The project will show how to carry uncertainty through a multi-step process while keeping model uncertainty, which more data reduces, separate from build-to-build scatter, which it does not. The decision method is the subject of the research and applies to other multi-step workflows, such as qualifying a new material or process.

Broader impacts. Undergraduates will make powder and prepare specimens, and a Ph.D. student will run the printing and microscopy campaigns, all gaining training in metal additive manufacturing and autonomous experimentation. We will release software, data, and models openly each year so other laboratories can apply the method to their own materials, printers, and tests.

Question for the program director. Does this topic fall within the scope of AMDO, or would it fit better in another program under NSF 26-515?